

Final Report: Global Pollution Analysis and Energy Recovery

1. Introduction

This project analyzes global pollution data to understand the relationship between pollution levels and the potential for energy recovery. Pollution sources such as industrial waste, CO₂ emissions, and other environmental indicators were examined to determine how they influence energy generation from waste-to-energy technologies.

The objective of this project is to preprocess pollution data, explore trends using visualizations, and build predictive models that can estimate energy recovery potential and classify pollution severity levels.

2. Dataset Description

The dataset contains environmental and economic indicators collected across multiple countries and years. It includes variables related to pollution levels, energy production, and population statistics.

Key features in the dataset include:

- Air Pollution Index
- Water Pollution Index
- Soil Pollution Index
- Industrial Waste (in tons)
- CO₂ Emissions (in MT)
- Renewable Energy (%)
- Plastic Waste Produced
- Energy Consumption Per Capita
- Population (in millions)
- GDP Per Capita (in USD)
- Energy Recovered (in GWh)

The dataset contains **200 records** and **13 attributes**.

3. Data Preprocessing

Before building machine learning models, several preprocessing steps were performed:

- Missing values were checked and handled appropriately.
- Categorical features such as **Country** were encoded using label encoding.
- Pollution indices were normalized using **StandardScaler**.
- Data types and statistical summaries were inspected to ensure data quality.

These steps ensured the dataset was ready for analysis and modeling.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns and relationships within the dataset.

Key visualizations included:

- **Correlation Heatmap** to identify relationships between pollution indicators and energy recovery.
- **Line Plot** showing trends in CO₂ emissions over time.
- **Box Plot** illustrating the distribution of industrial waste.
- **Scatter Plot** analyzing the relationship between industrial waste and energy recovery.

EDA helped reveal how pollution indicators interact with energy recovery and provided insights into potential predictive relationships.

5. Machine Learning Models

Linear Regression

A **Linear Regression model** was implemented to predict the amount of energy recovered based on pollution indicators.

Input Features

- Air Pollution Index
- CO₂ Emissions
- Industrial Waste

Target Variable

- Energy Recovered (in GWh)

Evaluation Metrics

- R^2 Score
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)

The model showed limited predictive accuracy, indicating that the relationship between the selected features and energy recovery may be weak or influenced by additional variables.

Logistic Regression

A **Logistic Regression model** was used to classify pollution severity into categories:

- Low
- Medium
- High

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The classification model achieved **high accuracy (~97%)**, demonstrating strong performance in categorizing pollution levels based on the available features.

6. Key Findings

The analysis revealed several important insights:

- Industrial waste and CO₂ emissions are significant indicators related to energy recovery potential.
- Pollution levels vary significantly across different countries and years.
- Machine learning models can effectively classify pollution severity levels.
- Waste-to-energy technologies have potential to convert environmental waste into usable energy.

7. Actionable Recommendations

Based on the analysis, the following recommendations can be made:

1. **Promote Waste-to-Energy Technologies**
Governments should invest in technologies that convert industrial waste into energy.
2. **Reduce Industrial Pollution**
Industries should adopt cleaner production techniques to reduce emissions and waste.
3. **Increase Renewable Energy Usage**
Expanding renewable energy sources can reduce reliance on polluting energy sources.
4. **Implement Environmental Monitoring Systems**
Data-driven monitoring systems can help track pollution levels and guide policy decisions.

8. Conclusion

This project demonstrates how data analysis and machine learning techniques can be used to understand environmental pollution trends and energy recovery potential. By leveraging pollution data, policymakers and environmental agencies can make informed decisions to reduce environmental impact while generating sustainable energy from waste.